

# 1 installation method

Step One: First, determine the installation location. This toy requires a vertical window that can be exposed to sunlight (horizontal skylights or attic windows with an incline are not suitable for installing this toy).

Step Two:  Be sure to wipe the glass clean first to prevent the adhesive hook from not sticking securely.

Step Three: First, attach the adhesive hooks to the product to ensure that the spacing between the hooks is appropriate.

Step Four: Peel off the adhesive backing from the hooks, then press the product firmly onto the glass to attach it.

Step Five: Remove the product from the hooks, press the hooks further to ensure they are securely attached, and then hang the product back on the hooks.

# 2 Initial Setup

Turn on the power switch of the product, use your phone or computer to find a hotspot named "rainbow" connect to this hotspot, and the password is 12345678

Then, using your phone's browser, open the URL 192.168.4.1 The interface will appear as shown in the image below. Follow the on-screen instructions to fill in your home Wi-Fi's name and password, the latitude, longitude, and altitude of the installation location, and the orientation of the window. Click 'Connect' to proceed.

### Rainbow Initialization



Does not collect any personal privacy from WiFi  
Designed by Whyso@BiliBili and Patent protection

Setup Interface

### Rainbow Initialization



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Filling Example

\*If you need to change the parameters, toggle the power switch to turn off and then turn on the device again, and repeat the above steps

## Parameter Explanation And Tips

1:The target Wi-Fi for connection only supports 2.4G; attempting to connect to a 5G Wi-Fi hotspot will fail. When setting up a mobile hotspot on your phone, please select the 'Best Compatibility' option."

2:Verify the Wi-Fi password to ensure it is correct. Public Wi-Fi hotspots that require additional verification to access the internet after connecting are not suitable for this toy.

### 3: weather forecast source private key

If you do not use the weather forecast feature, it can be set to 'OFF'. Weather is a third-party weather forecast information service provider. Users can register for a free account using their personal phone number to obtain a private key. A registration video tutorial is available; scan the QR code below to access and watch the instructions. You can also leave it blank, in which case the built-in default private key will be used. However, this may result in weather forecast updates failing due to excessive usage by many users.

### 4: latitude and longitude data

The format for entering latitude and longitude data should be in decimal format, with 5 to 6 decimal places to provide precise location information. If the data you find is in degrees-minutes-seconds (DMS) format, for example 30°20'56". You can convert to decimal format using the formula: decimal format = D + M/60 + S/3600.

A more convenient way is to look up the latitude and longitude online. The QR code for the lookup link is provided below.



Weather API Keys



Latitude&Longitude Query

## 5: Altitude

You can check the altitude using a map APP or by installing a third-party compass APP. The altitude value does not need to be precise; an error of around a hundred meters will generally have no significant impact.

### Hide Additional Parameters

After the altitude value, an Easter egg setting can be appended for users with special application needs to explore. Typically, there is no need to add this. The format and instructions for the additional settings are as follows:

\*Brackets are necessary in optional parameters, UTC/ADJ/NTP are case-sensitive

80.356481[UTC-5][ADJ3.0][NTPtime.google.com]

Altitude Value (Required) optional parameters Custom Time Zone optional parameters Fine-tune the coordinate system optional parameters Custom NTP time server

To address step motor gear errors, a setting has been added to allow fine-tuning of the coordinate system.

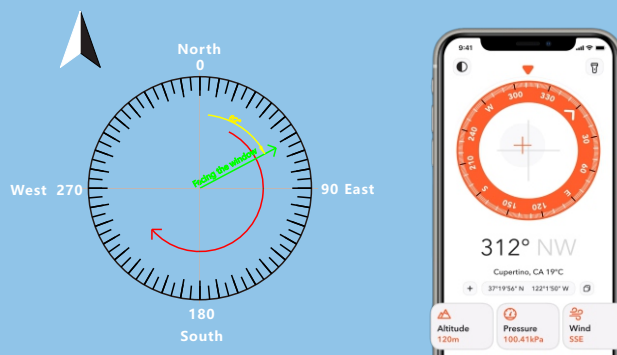
To make the rainbow smaller and brighter (more focused beam), you can slightly adjust the coordinates in the negative direction. For example, enter: 80.35[ADJ-3.0]

To make the rainbow area larger (more spread out), you can slightly adjust the coordinates in the positive direction. For example, enter: 80.35[ADJ3.0]

Generally, the fine-tuning angle should be within  $\pm 6$  degrees. Over-adjusting may result in the inability

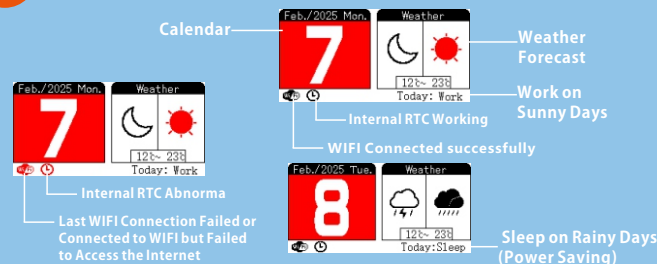
## 6: Window orientation

Window orientation is a pure numeric value, with true north being 0 degrees, and following the sequence of North-East-South-West-North, it corresponds to 0-360 degrees, as shown in the image below. You can use the reading from your phone's built-in compass APP when pointing it towards the window to get the window orientation value.



The latitude, longitude, altitude, and window orientation parameters can actually be obtained from the built-in 'Compass' app on your phone. When obtaining these values, please hold your phone flat and facing the window directly. (Note: The built-in 'Compass' app on Apple devices no longer displays latitude, longitude, and altitude information starting from iOS 14. You can download a third-party 'Compass' app from the App Store to get this information.)

# 3 on-screen information



# 4 Precautions

- When attaching to indoor glass, please clean the glass in advance to avoid dust or oil stains that may cause the adhesive to fail.
- Do not manually rotate the product with external force, as this may cause internal gears to break and damage the product.
- Do not install the product outdoors at high levels to prevent accidental falls that could cause injury to others.
- When the battery is depleted, please use a Type-C USB cable to charge.
- This product is not suitable for use or play by children under 9 years old.
- This product does not have the function of focusing light and will not cause fire hazards due to sunlight concentration and temperature rise.



Communication and Q&A

